Apply to be a Science Corps Fellow - Entry #3347

Name

Swati Yadav

Email

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Gender

Female

Residence: City (State), Country

Kodigehalli

Nationality

Indian

When did you or do you expect to complete your PhD?

September 2024

Please tell us briefly about your university education (institution, city, country, year)

I hold an Integrated BS-MS degree in Chemical Sciences from the University of Hyderabad, where I completed my studies in 2017. During this time, I built a strong foundation in organic, inorganic, and physical chemistry. My MS dissertation focused on computational analysis of infrared spectra.
Following this, I was selected for the PhD program at the National Centre for Biological Sciences (NCBS), Bangalore, which I completed in 2024. At NCBS, I specialized in structural biology, focusing on the molecular mechanisms of protein function. My research involved studying both membrane and soluble proteins using cryo-electron microscopy (cryo-EM), with an emphasis on eukaryotic transporters, mechanosensitive channels, and bacterial restriction enzymes.

Why would you like to be a Science Corps Fellow?

I want to be a Science Corps fellow because I'm looking for a way to give back to society and find meaning in the work I do. After years of focusing on research, I feel a strong pull to step outside the lab and use what I've learned to make a direct impact. I'm especially drawn to the idea of contributing to science education in under-resourced settings and helping others build confidence in learning and exploring science. I see this as a chance to connect with people, share knowledge, and grow in ways that research alone doesn't offer.

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

Alongside my academic training, I have actively sought out opportunities that would help me grow as a Science Corps Fellow. During my PhD, I had the chance to mentor junior lab members, which involved leading practical

sessions on cryo-EM techniques. This experience not only gave me a chance to share my knowledge but also helped me hone my communication skills, especially when explaining complex technical concepts. Additionally, I served as a workshop instructor for graduate students, where I taught the fundamentals of cryo-EM data processing. This role further developed my ability to convey intricate ideas in a clear, accessible way.
Beyond my research work, I participated in several institute open days, where I interacted with students of all ages through science demonstrations and poster presentations. These experiences allowed me to engage with a wider audience, and they reinforced my passion for sharing science beyond the academic community. I found it both enjoyable and rewarding to make science accessible to everyone, regardless of their background.
Teaching, learning from others, and exploring new ways to contribute are aspects of science that I find truly fulfilling.

Where do you imagine yourself professionally in the future? Check all that apply.

Scientist in academia
In science education
In science outreach
Other

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

In the next two years, I see myself learning, diversifying, and figuring out the best way to contribute to and share science. I want to explore different avenues and find where I can make the most impact. In ten years, I envision myself settled in a role, whether as a lab lead, science communicator, or educator, focusing on fundamental science with translational value. One concept that resonates with me is "frugal science"—where observation, hypothesis-driven testing, and method building are prioritized to get results, rather than just following a standard pipeline. This approach is something I deeply believe in and aim to incorporate into my work.

Please upload your CV (.pdf)

CV_Swati_Yadav.pdf

How did you find out about Science Corps?

An email list in your institution

Science Corps